Surya Prakash Singh

Full Stack Software Development Engineer II

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Professional Summary

Full Stack Software Development Engineer II with **3 years** of production experience building revenue-critical, real-time systems in healthcare tech, end-to-end across **React.js/React Native** frontends, **Node.js** backends, and **PostgreSQL/MongoDB** data layers. Sole owner of the **Cashless Dental module** at Visit Health, the platform's primary revenue-generating product, processing **15,000+ monthly appointments** end-to-end across UI, payment flows, wallet deduction, OTP verification, and multi-actor integrations. Built a **Slack-inspired real-time messaging platform** end-to-end (React Native mobile, React.js web, Node.js/Socket.io backend) supporting **2,500+ concurrent users** with real-time message delivery. Specialized in **TypeScript, component-based architecture, REST APIs**, and event-driven systems.

Work Experience

Visit Health Bengaluru, Karnataka, India

Software Development Engineer II Sep 2025 – Present

- Led end-to-end development of the **Cashless Dental module**, the platform's primary revenue-generating product, owning the full stack including appointment booking, wallet deduction, payment flows, OTP verification, and partner integrations across a multi-actor system serving patients, doctors, clinics, and ops teams, processing **15,000+ monthly appointments** end-to-end.
- Reduced support tickets by **50%** by engineering proactive prescription approval/rejection notifications, eliminating manual intervention across the prescription lifecycle.
- Built **React.js booking, wallet, and OTP verification screens** for the Cashless Dental module, cutting booking-flow completion time by **40%** and reducing OTP-related drop-off by **50%** through improved error and retry handling.
- Built a **multi-channel comms dashboard** (Email/SMS/WhatsApp/Push) tracking delivery status across Dental and Digitization service lines, cutting ops' manual status-check time by **80%**.
- Integrated **FitPass partner APIs** into the Visit Health app enabling gym discovery, session booking, and post-workout reviews, supporting **10,000+ monthly bookings** through the platform.
- Integrated **Aditya Birla Health Insurance APIs** to automate health checkup booking notifications, reducing manual ops overhead by **70%**.
- Architected a **doctor-patient call management system** tracking interaction history, missed attempts, and call duration analytics, reducing missed-follow-up rate by **80%** for the ops team.
- Applied **Spec-Driven Development (SDD)** and an **AI-augmented development lifecycle (AI-DLC)** to accelerate feature delivery through AI-assisted spec authoring and code generation.

Diatoz Solutions Private Limited Bengaluru,Karnataka, India

Software Development Engineer Jan 2024 – Aug 2025

- Engineered a **Slack-inspired real-time messaging platform** from scratch as a greenfield project, Node.js, Socket.io, and MongoDB backend supporting direct, group, and channel messaging for **5,000+ users** with real-time delivery and status sync.
- Built **React.js web and React Native mobile clients** with hooks-based state management, handling cross-platform state sync for message history and notifications.
- Designed **MongoDB schemas** optimized for high-throughput querying, supporting **150,000+ messages/day** across **500+ concurrent channels** with read/unread tracking and real-time notification.
- Built a **role-based channel management system** with granular permission controls (Admin, Maintainer, Member) and conditional UI rendering per role across **500+ channels**.
- Developed **typing indicators and online presence tracking** over WebSockets for **2,500+** concurrent connections, plus polling, pinned channels, and Google Meet link generation for instant video calls.

Diatoz Solutions Private Limited Bengaluru,Karnataka, India

Software Development Engineer Intern Aug 2023 – Dec 2023

- Developed a mentorship platform using **Next.js**, resulting in **150+ monthly student enrollments**.

Skills

Languages: JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, React Native, Redux Toolkit, Context API, HTML, CSS

Backend: Node.js, Express.js, NestJS, REST APIs, Socket.io, WebSockets

AI & LLM Engineering: OpenAI API, Anthropic Claude API, LangChain.js, RAG, Vector Databases (pgvector, Pinecone), Embeddings, Prompt Engineering, Function Calling / Tool Use, MCP (Model Context Protocol), AI Agents, Streaming LLM Responses

Databases: MongoDB, PostgreSQL, MySQL

DevOps & Infrastructure: Docker, Linux, AWS

Concepts: Real-Time Systems, Event-Driven Architecture, Scalable System Design, SDD, AI-DLC